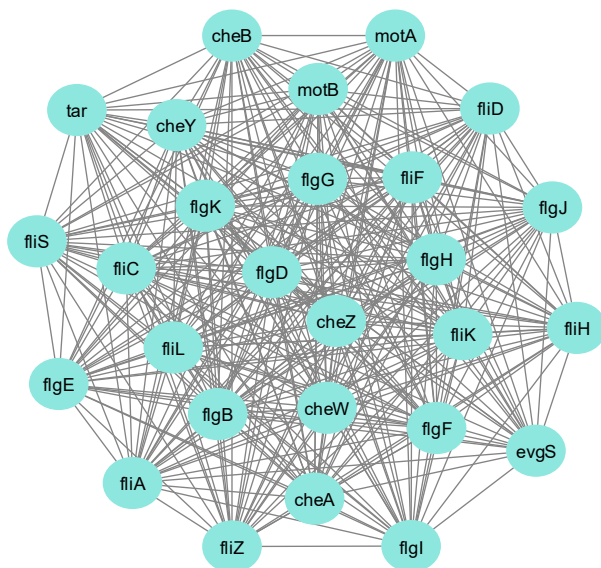


Functional modules identified by MCODE analysis.

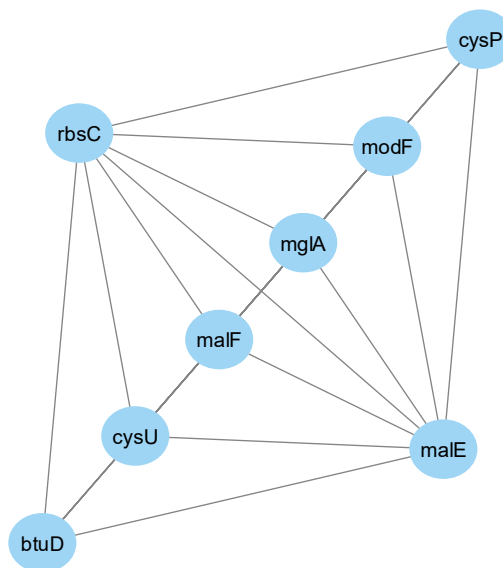
A Cluster 1 – Flagellar Assembly, Motility and Chemotaxis



**MCODE score: 26.538 | 27
nodes | 345 edges**

Flagellar structural proteins and chemotaxis-associated factors involved in bacterial motility, environmental sensing, and directional movement.

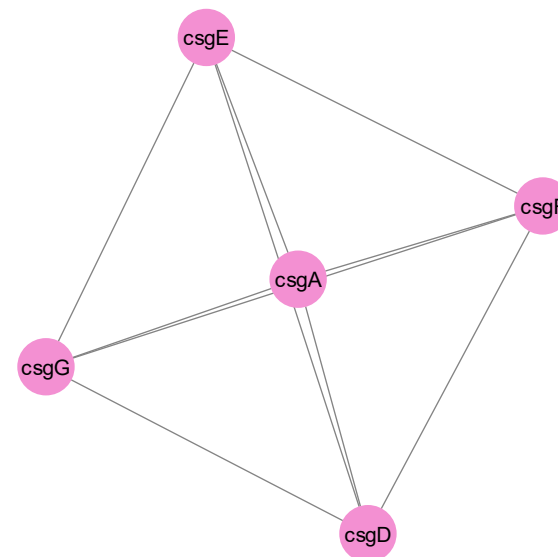
B Cluster 3 – Nutrient Transport and Uptake



**MCODE score: 8.000 | 8 nodes |
28 edges**

Transport and uptake proteins involved in nutrient acquisition and cellular adaptation to changing environmental conditions.

C Cluster 4 – Curli and Biofilm Regulation



MCODE score: 5.000 | 5 nodes | 10 edges

Curli structural and regulatory proteins involved in biofilm formation, surface adhesion, and multicellular bacterial adaptation.

MCODE: Molecular Complex Detection

- **fli*, flg***: Flagellar assembly

- **che*, tar:** Chemotaxis

- **motA, motB:** Flagellar motor

cysP, cysU, malE, malF, modF, btuD, mglA, rbsC: Transport

- **csgA, csgE, csgF, csgG:** Curli assembly

- **csgD**: Biofilm regulation